

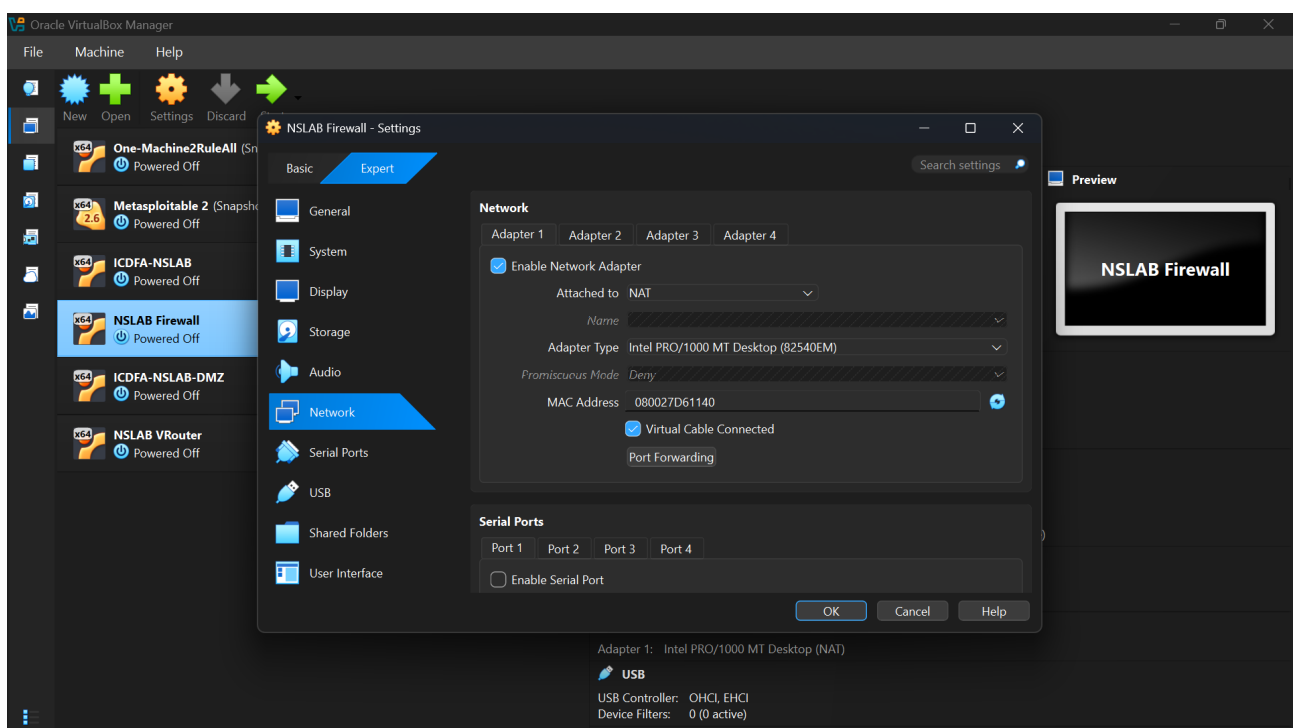
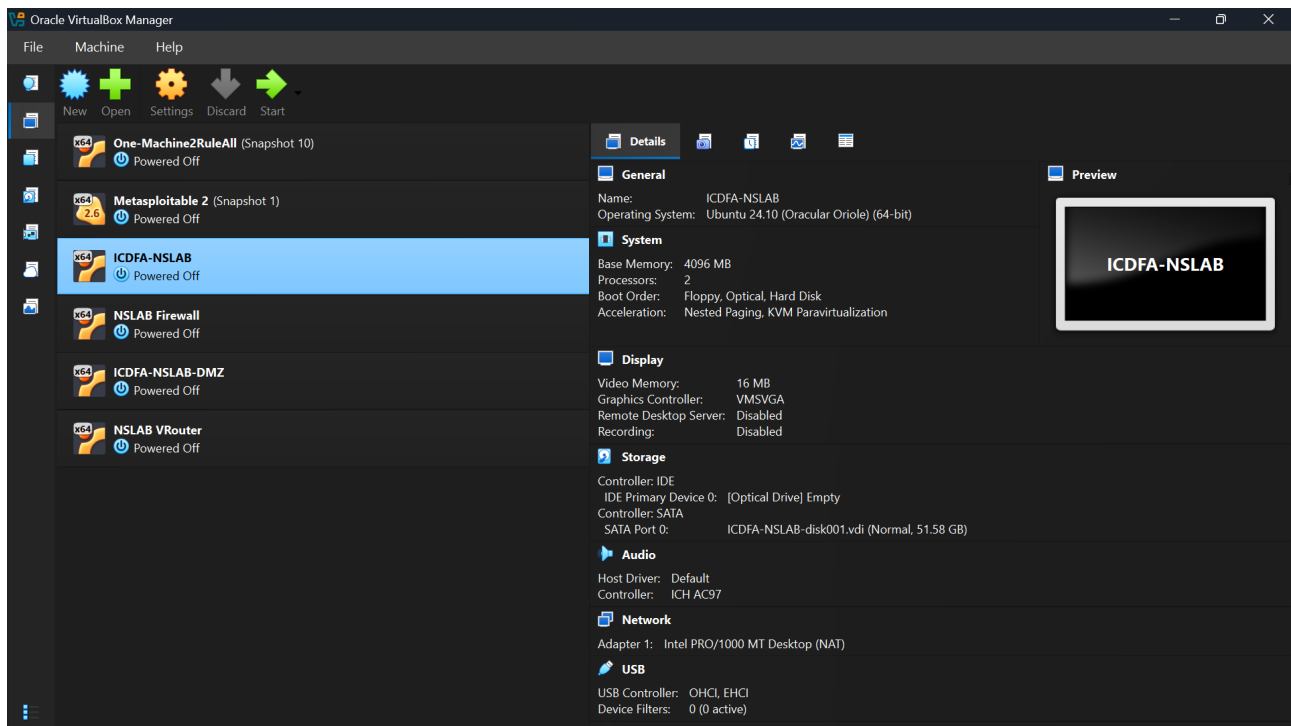
**Full name: Adeleke Oluwapelumi Faruq**

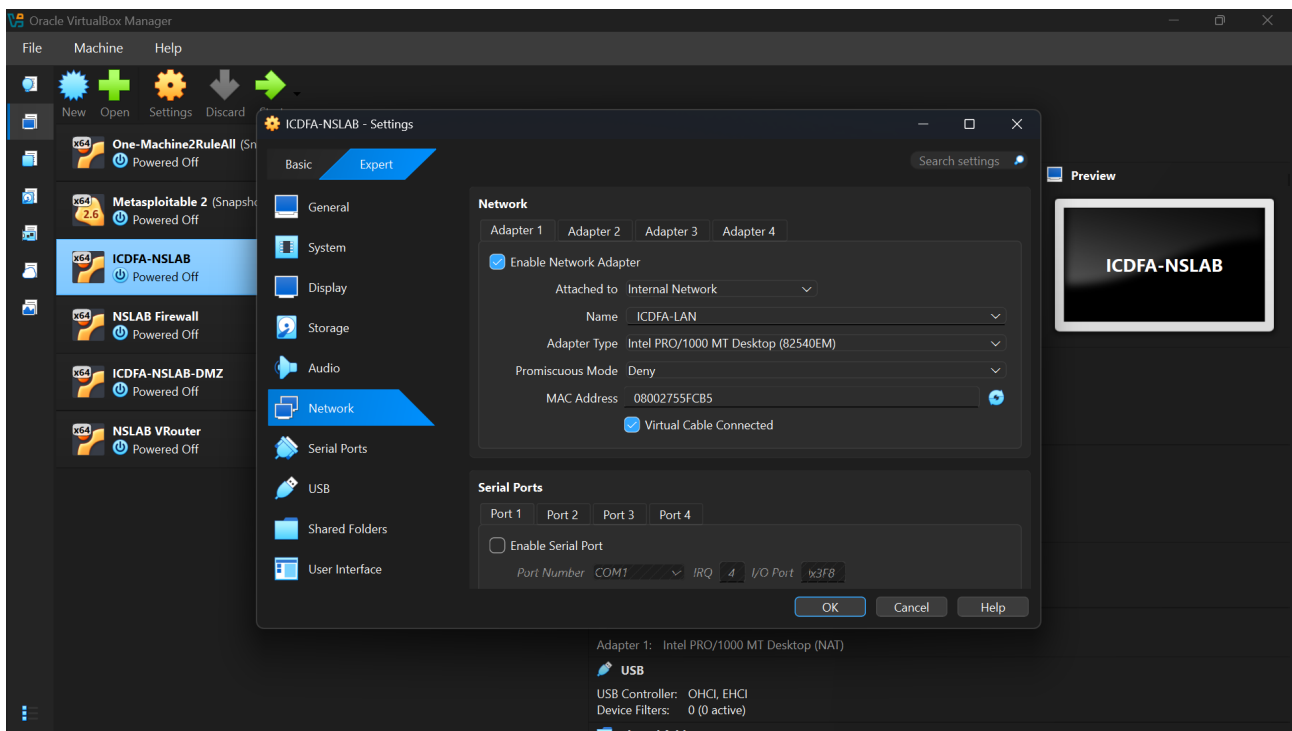
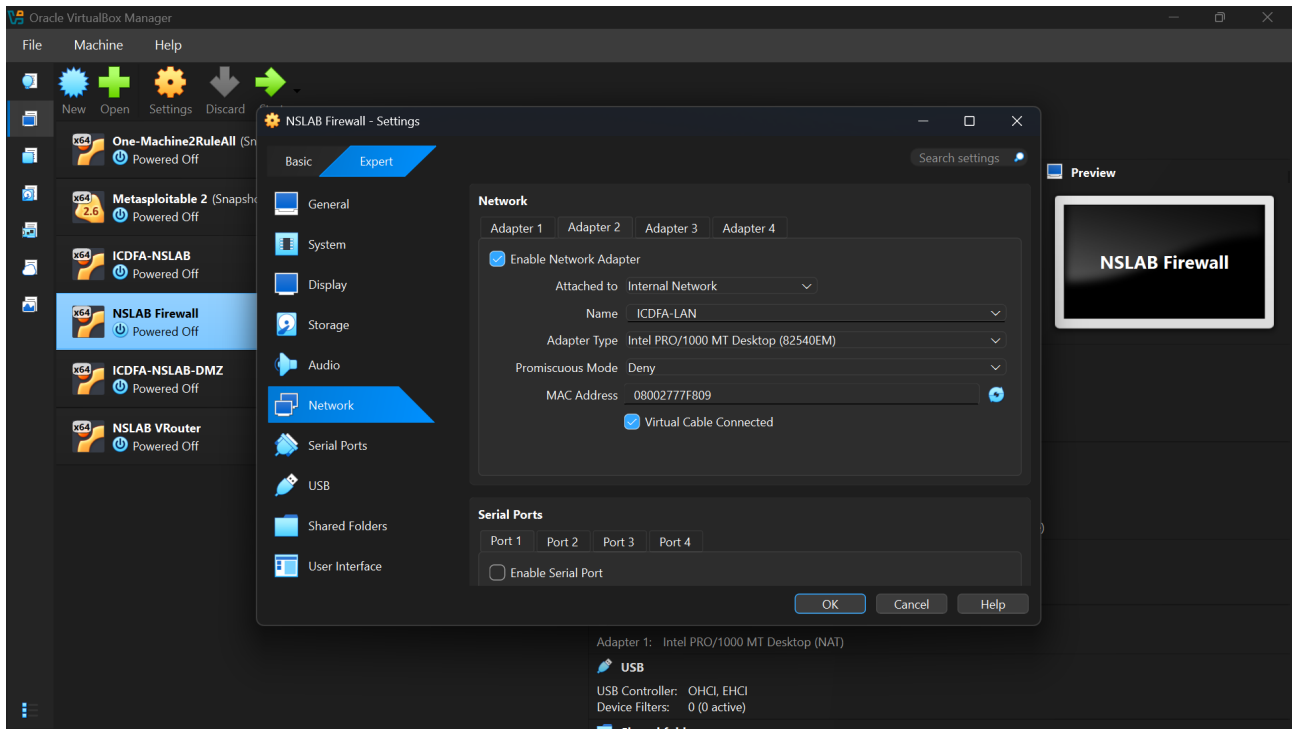
**Registration number: C11/26/FCDF/17167**

**Linux distribution / version: Ubuntu 24.10 (Oracular Oriole) (64-bit)**

**VM hostname: Oracle VirtualBox**

## Part A:





## Part B

```
NSLAB Firewall [Running] - Oracle VirtualBox
File Machine View Input Devices Help
: Forums: https://forum.opnsense.org/ : : :
: Code: https://github.com/opnsense : : : : :
: Reddit: https://reddit.com/r/opnsense : : : : :
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*** icdfa-nslab-firewall-v1.icdfa.test: OPNsense 26.7 (amd64) ***

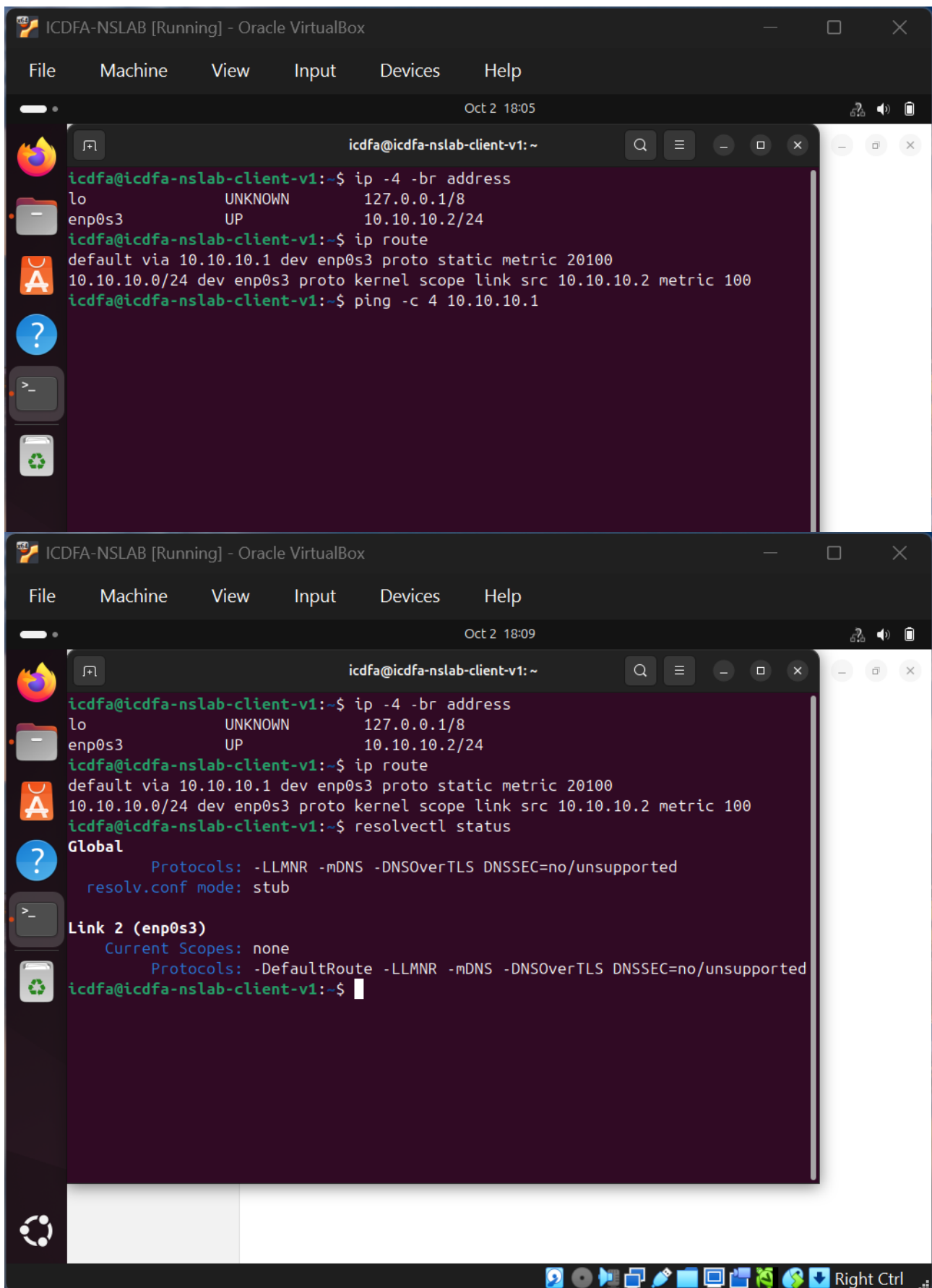
LAN (em0) -> v4: 10.10.10.1/24
WAN (em1) ->

HTTPS: SHA256 E8 0B 30 3E D5 89 EB 63 A0 D8 83 2F E6 38 4C 22
      82 2C 5A A4 39 D5 45 54 57 50 A0 73 58 E0 46 90
SSH: SHA256 K10Lt4mW484dUmuPy37tD714RA0De5LKjSU/+bKB1Tw (ECDSA)
SSH: SHA256 5wvyP8AD885sP0fHz7CgfNStFMhPw42x23cBxco4 (ED25519)
SSH: SHA256 6e/0XrjSdfL/Hsy8atZw0IPQGsb1P6zFfuJ/FZ50d+A (RSA)

0) Logout
1) Assign interfaces
2) Set interface IP address
3) Reset the root password
4) Reset to factory defaults
5) Power off system
6) Reboot system
7) Ping host
8) Shell
9) pfTop
10) Firewall log
11) Reload all services
12) Update from console
13) Restore a backup

Enter an option: █
```

## Part C



The image shows two screenshots of a terminal window running inside an Oracle VM VirtualBox. The window title is "ICDFA-NSLAB [Running] - Oracle VirtualBox". The terminal prompt is "icdfa@icdfa-nslab-client-v1: ~".

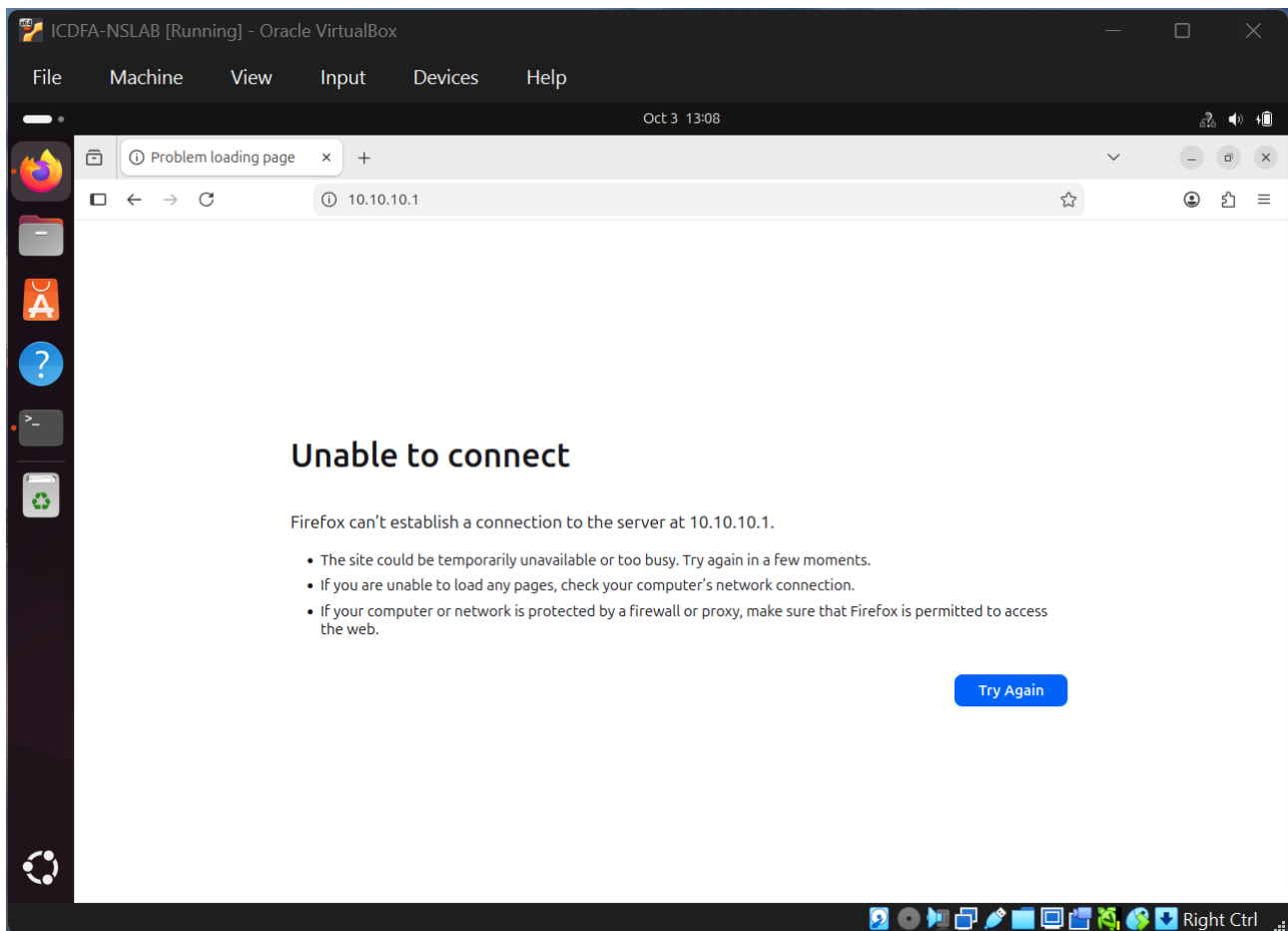
**Top Screenshot (Oct 2 18:05):**

```
icdfa@icdfa-nslab-client-v1:~$ ip -4 -br address
lo                UNKNOWN      127.0.0.1/8
enp0s3            UP            10.10.10.2/24
icdfa@icdfa-nslab-client-v1:~$ ip route
default via 10.10.10.1 dev enp0s3 proto static metric 20100
10.10.10.0/24 dev enp0s3 proto kernel scope link src 10.10.10.2 metric 100
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 10.10.10.1
```

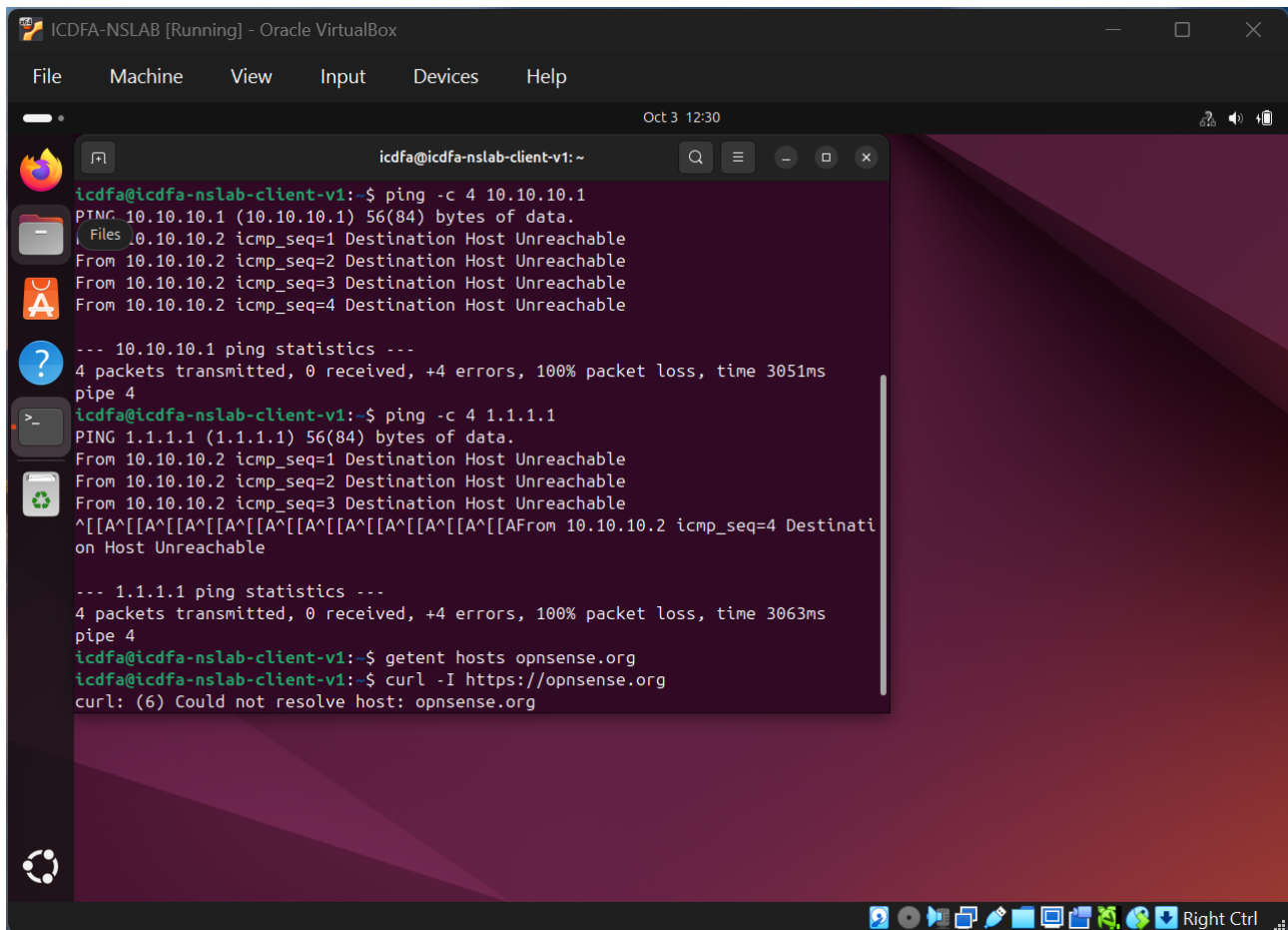
**Bottom Screenshot (Oct 2 18:09):**

```
icdfa@icdfa-nslab-client-v1:~$ ip -4 -br address
lo                UNKNOWN      127.0.0.1/8
enp0s3            UP            10.10.10.2/24
icdfa@icdfa-nslab-client-v1:~$ ip route
default via 10.10.10.1 dev enp0s3 proto static metric 20100
10.10.10.0/24 dev enp0s3 proto kernel scope link src 10.10.10.2 metric 100
icdfa@icdfa-nslab-client-v1:~$ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    resolv.conf mode: stub

Link 2 (enp0s3)
    Current Scopes: none
    Protocols: -DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
icdfa@icdfa-nslab-client-v1:~$
```



## Part D

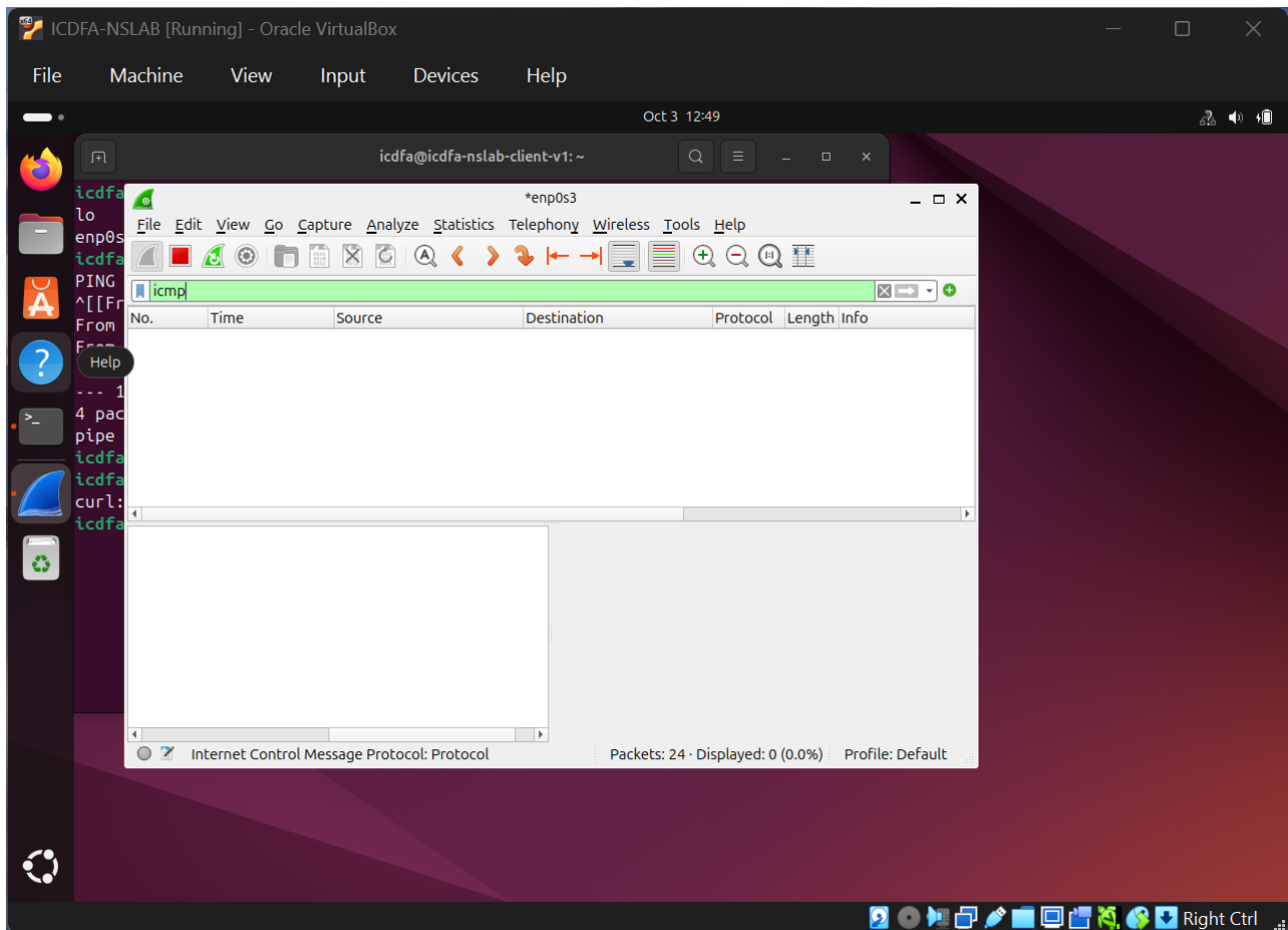
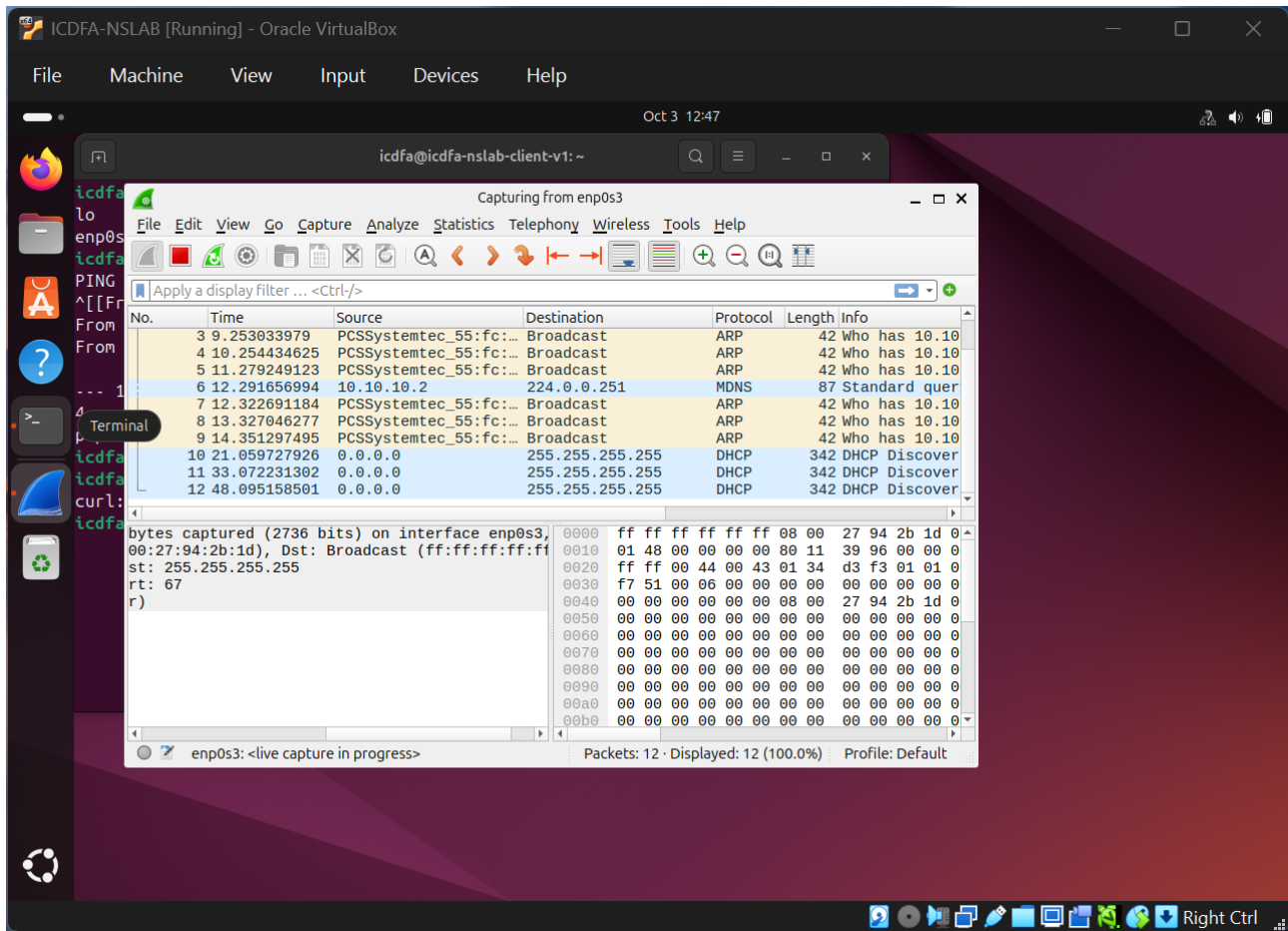


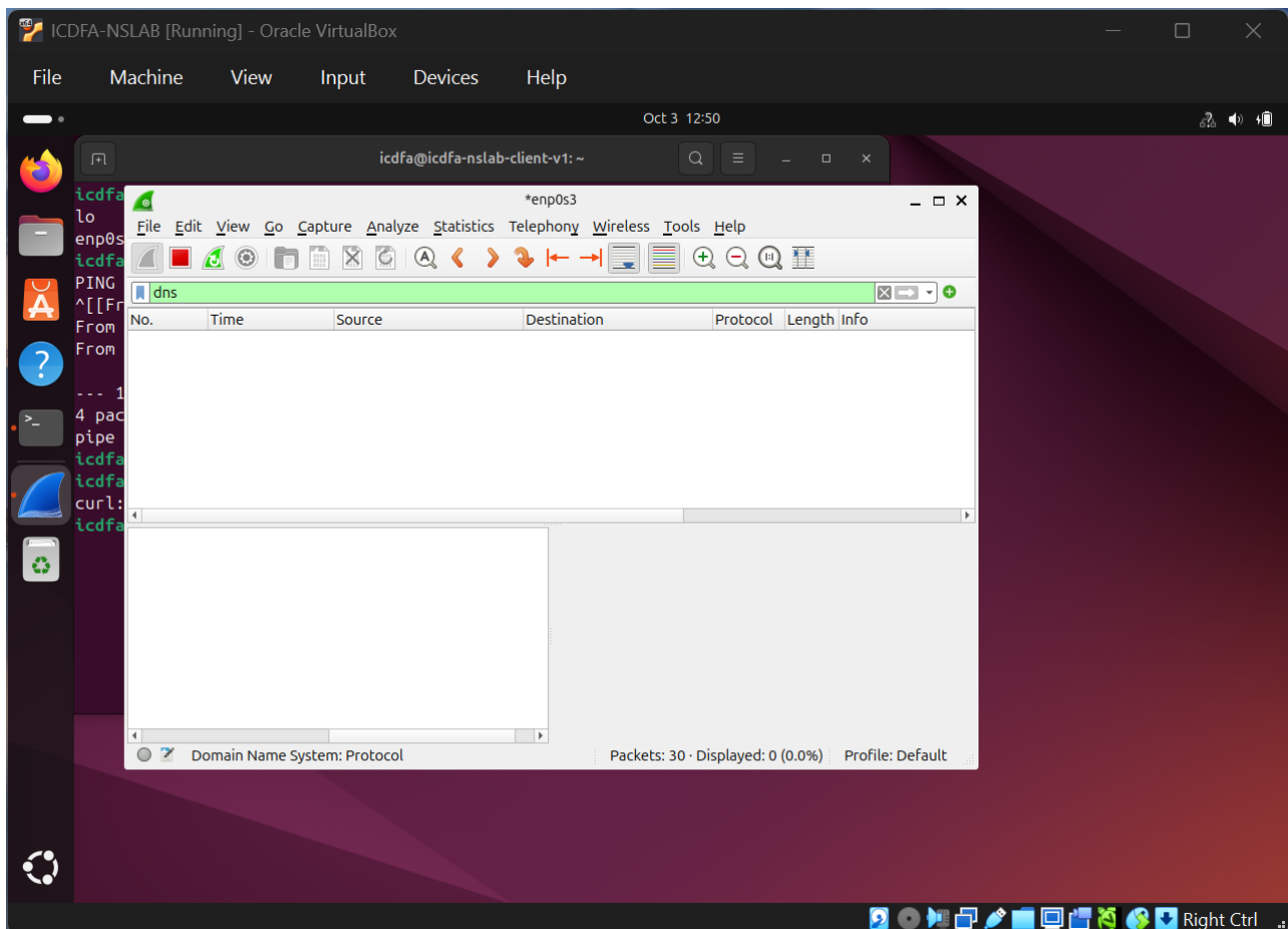
```
ICDFA-NSLAB [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Oct 3 12:30
icdfa@icdfa-nsrab-client-v1:~
icdfa@icdfa-nsrab-client-v1:~$ ping -c 4 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
From 10.10.10.2 icmp_seq=1 Destination Host Unreachable
From 10.10.10.2 icmp_seq=2 Destination Host Unreachable
From 10.10.10.2 icmp_seq=3 Destination Host Unreachable
From 10.10.10.2 icmp_seq=4 Destination Host Unreachable

--- 10.10.10.1 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3051ms
pipe 4
icdfa@icdfa-nsrab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
From 10.10.10.2 icmp_seq=1 Destination Host Unreachable
From 10.10.10.2 icmp_seq=2 Destination Host Unreachable
From 10.10.10.2 icmp_seq=3 Destination Host Unreachable
From 10.10.10.2 icmp_seq=4 Destination Host Unreachable

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 0 received, +4 errors, 100% packet loss, time 3063ms
pipe 4
icdfa@icdfa-nsrab-client-v1:~$ getent hosts opnsense.org
icdfa@icdfa-nsrab-client-v1:~$ curl -I https://opnsense.org
curl: (6) Could not resolve host: opnsense.org
```

## Part E





## Questions

1. What is the difference between the OPNsense WAN and LAN interfaces?  
 WAN: The untrusted, external facing interface connected to VirtualBox NAT.  
 LAN: The trusted, internal-facing interface connected to the ICDFA-LAN network.
2. Why must both internal adapters use the same VirtualBox network name? In VirtualBox, the Internal Network name functions as a virtual broadcast domain. Only VMs assigned the exact, name connect to the same virtual switch fabric and communicate at Layer 2.
3. What information does the default route provide to Ubuntu? The default route specifies the next-hop IP address to which Ubuntu must forward any packet whose destination IP address does not match a directly connected local subnet.
4. Which packet exchange allows Ubuntu to learn the firewall MAC address? The ARP exchange: Ubuntu broadcasts an ARP Request and the firewall replies with a unicast ARP Reply containing its Layer 2 MAC address.
5. Why does a successful ping to 1.1.1.1 not automatically prove that DNS is working? ping 1.1.1.1 bypasses domain name resolution completely by transmitting ICMP packets directly to a Layer 3 IP address. It only validates IP routing, default gateway reachability, and outbound NAT.